

Mechanical Performance and Mix Optimization of M50 Concrete Incorporating Metakaolin and Waste Foundry Sand

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Abstract

In this study, the joint effect of metakaolin (MK) as a partial cement replacement and waste foundry sand (WFS) as a partial natural-sand replacement is investigated on the workability and mechanical strength of M50 concrete mixtures. A 6 × 5 experimental matrix was adopted, comprising MK contents of 0, 3, 6, 9, 12 and 15% and WFS contents of 0, 10, 20, 30 and 40%. The water to binder ratio was kept constant at 0.35. All 30 combinations were tested for slump and compressive strength after 7 and 28 days and split tensile and flexural strengths were also screened using separate series of one factor. This increase in water requirement of the finer constituents is confirmed by the reduction in slump from 125 mm for control mixture to 90 mm at the highest combined replacement level. The maximum 28 days compressive strength of 73.58 MPa occurred at 20% WFS and 12% MK (an increase of 22.94% over the control strength of 59.85 MPa). The 12% MK and 30% WFS mixture recorded the absolute maximum strengths of 54.08 MPa at 7 days and 56.81 MPa at 28 days, respectively, compared to the control. The one factor screening series resulted in 12% MK having the highest 28-day split tensile (5.12 MPa) and flexural (8.68 MPa) strengths, while 15% MK had a significant decrease. Based on the results, it can be concluded that the most balanced mixture was 12% MK and 20% WFS which had the maximum 28-day compressive strength of 73.58 MPa and a workable slump of 118 mm. The results show that the combination of these two replacements can generate high strength concrete, while utilizing foundry waste and minimizing the consumption of natural sand.

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1. INTRODUCTION

Concrete production depends heavily on Portland cement and natural aggregates. This has resulted in increased interest in supplementary cementitious materials and alternative aggregates, due to the environmental impacts of the clinker production and the continued mining of river sand. Metakaolin is a highly reactive aluminosilicate which is produced by controlled calcination of kaolinitic clay. The fine particles act as a filler and produce pozzolanic reaction with calcium hydroxide to form extra calcium silicate hydrate and calcium aluminosilicate hydrate phases. These mechanisms may enhance the state of microstructures, increase the

strength of the interfacial transition zone and enhance the early and later-age mechanical performance [1–5]. Foundry waste is the sand which is no longer usable for the casting and moulding of metal. Generally silica rich, it is finer than conventional natural sand, and may contain clay, carbonaceous matter or chemical binders according to foundry process. The studies conducted in the past have indicated that moderate replacement of WFS could retain or enhance the strength of the cementitious composite because particle packing, but excessive replacement may increase the water requirement and decrease the workability, and may weaken the paste-aggregate bond [6-12]. The replacement level for optimum properties seems to vary in the literature, but it is often suggested that 10-30% be used for structural concrete. Technically appealing is the combination of MK and WFS, as each material can be used to address a different component of concrete: MK can be used to reduce the amount of Portland-cement required, and WFS can be used to decrease the need for natural-sand and to replace an industrial by-product normally disposed of. Siddique and Kadri [13] has shown that the addition of 20% foundry sand along with 5–15% MK can enhance the properties of sand in near surface zone. The combined effect of a wider replacement matrix and its effect on the strength development of high strength concrete, however, is less documented, especially using a low water-to-binder ratio. The present work explores M50 concrete with 6×5 matrix of MK and WFS contents. Specific objectives to achieve were: (i) quantify the changes in slump due to the two replacement materials; (ii) determine the split tensile and flexural strengths for all combinations; (iii) screen the individual effects of WFS and MK on split tensile and flexural strengths; (iv) find a balanced optimum based on strength and workability; and (v) give an indicative material-cost comparison using the rates adopted in the project. The study is limited to mechanical properties at 7 and 28 days, and did not include durability, leaching and microstructural characterization.

2. MATERIALS AND METHODS

2.1 Constituent materials

The main binder used was ordinary Portland cement (53 grade). Coarse aggregate (nominal maximum size 20 mm) used had a specific gravity of 2.65, water absorption of 2.94% and fineness modulus of 6.87. The natural sand available locally had a specific gravity of 2.50, water absorption of 4.0% and a fineness modulus of 3.17. The waste foundry sand was gathered from a foundry in Chhatrapati Sambhajinagar, Maharashtra, India. Material was screened prior to use, and had a measured specific gravity of 2.38. The metakaolin was provided in a dry powder form

that was white in color and had a specific gravity of 2.30. Manufacturer data indicated 50–55% SiO₂ and 38–42% Al₂O₃. Low-water-binder-ratio mixtures were obtained by using a sulphonated naphthalene polymer superplasticizer (Conplast SP 430) having a specific gravity of 1.220–1.225 and a pH of 7–9. Curing and mixing were done with potable water.

Table 1. Principal characteristics of the constituent materials.

Material	Specific gravity	Water absorption (%)	Fineness modulus	Other reported property
OPC 53 grade	3.15	-	-	Fineness 3%; consistency 29%; setting 135/288 min
Natural sand	2.50	4.00	3.17	Locally sourced
Coarse aggregate	2.65	2.94	6.87	Crushed; 20 mm maximum size
Waste foundry sand	2.38	-	-	Screened foundry by-product
Metakaolin	2.30	-	-	SiO ₂ 50–55%; Al ₂ O ₃ 38–42%
Superplasticizer	1.220–1.225	-	-	Sulphonated naphthalene polymer; pH 7–9

2.2 Mix design and experimental matrix

The proportions of the reference concrete were designed based on IS 10262:2019 and IS 456:2000 [14,15] for M50 grade. The desired strength was 58.25 MPa and water-to-binder ratio was 0.35. Metakaolin replaced cement by mass at 0, 3, 6, 9, 12 and 15%, while WFS replaced natural sand by mass at 0, 10, 20, 30 and 40%. This produced 30 combinations. The dose of superplasticizer was about 1.2% of the cementitious material and small quantities of water and aggregate were adjusted due to absolute volume calculations depending on the mix. The amount of cement, water, natural sand, and coarse aggregate used for the control mixture were 434.3 kg/m³, 152.0 kg/m³, 574.2 kg/m³ and 1234.9 kg/m³, respectively. Later, the mixture that was selected as the balanced optimum mixture contained 379.6 kg/m³ of cement, 51.8 kg/m³ of MK, 113.0 kg/m³ of WFS, 452.2 kg/m³ of coarse aggregate and 1228.6 kg/m³ of water, and 151.0 kg/m³ of water. This resulted in a replacement that reduced the amount of Portland cement used by 54.7 kg/m³ compared to the control, and the natural sand used was decreased by 122.0 kg/m³.

Table 2. Representative quantities for the control and balanced optimum mixtures.

Mix	MK (%)	WFS (%)	Water	Cement	MK	Natural sand	WFS	Coarse aggregate
Control	0	0	152.0	434.3	0	574.2	0	1234.9
Balanced optimum	12	20	151.0	379.6	51.8	452.2	113.0	1228.6

Note: Quantities are in kg/m³ except replacement percentages.

2.3 Specimen preparation and testing

The dry constituents were weighed separately and blended until visually uniform. Water containing the required superplasticizer dosage was then added gradually, followed by continued mixing to achieve a homogeneous mixture. Slump was measured for each batch using the standard slump-cone procedure [17]. Specimens were cast in cleaned and oiled steel moulds, compacted in layers, finished with a trowel and demoulded after approximately 24 h. The specimens were subsequently water-cured until testing.

Compressive strength was determined on 150 × 150 × 150 mm cubes at 7 and 28 days using a compression testing machine in accordance with IS 516:1959 [16]. Three cubes were tested for each age and mixture, and the reported values are the arithmetic means. Split tensile strength was measured on 150 × 300 mm cylinders. Flexural strength was measured on 100 × 100 × 500 mm beams over a 400 mm effective span under centre-point loading. The governing expressions were:

$$f_c = P / A$$

$$f_t = 2P / (\pi LD)$$

$$f_r = 3PL / (2bd^2)$$

where f_c is cube compressive strength, f_t is split tensile strength, f_r is flexural strength, P is the failure load, A is the loaded area, L is specimen length or beam span as applicable, D is cylinder diameter, and b and d are the beam width and depth. The compressive-strength program used the complete 30-mixture matrix. Split tensile and flexural strengths were evaluated through separate one-factor screening series for WFS and MK; consequently, those results should not be interpreted as a combined factorial response.

3. RESULTS AND DISCUSSION

3.1 Workability

The slump values ranged from 125 to 90 mm. At 0% MK, increasing WFS from 0 to 40% reduced slump from 125 to 110 mm. At 15% MK, the same increase in WFS reduced slump from 110 to 90 mm. The reduction was caused by the combined influence of the high specific surface area of MK and the finer, more absorptive fraction associated with WFS. These particles increase the paste demand and reduce the amount of free water available for flow. Similar reductions in workability with increasing WFS and MK have been reported in earlier studies [1,8,10,12].

The 12% MK–20% WFS mixture retained a slump of 118 mm, only 5.6% below the control. This indicates that, with adequate superplasticizer, the balanced optimum remained practically workable. By contrast, mixtures containing 15% MK and 30–40% WFS had slump values of 95 and 90 mm, respectively, which may impair compaction in densely reinforced members unless the admixture dosage or mixing procedure is further optimized.

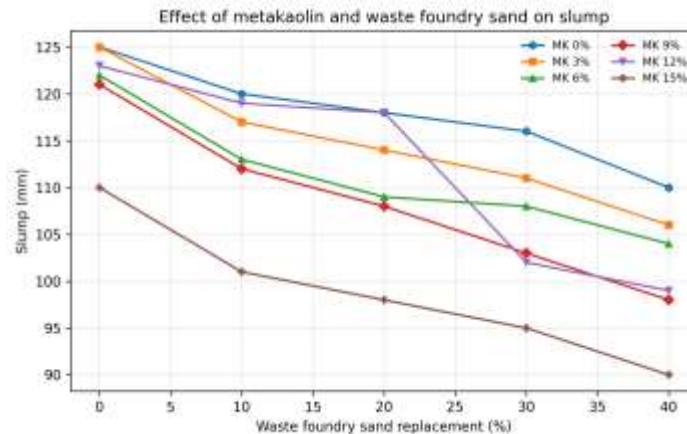


Figure 1. Slump variation with waste foundry sand at different metakaolin contents.

Table 3. Slump matrix for the 30 concrete mixtures (mm).

MK (%)	WFS 0%	WFS 10%	WFS 20%	WFS 30%	WFS 40%
0	125	120	118	116	110
3	125	117	114	111	106
6	122	113	109	108	104
9	121	112	108	103	98
12	123	119	118	102	99
15	110	101	98	95	90

3.2 Compressive strength

All mixtures exceeded the 50 MPa characteristic strength at 28 days except none; even the lowest 28-day mean, 56.80 MPa at 15% MK and 0% WFS, remained above the nominal grade requirement. However, the response was strongly non-linear. Moderate WFS replacement generally improved strength, whereas the best WFS level varied between 10 and 30% depending on MK content. This source-dependent behaviour is consistent with the dual role of WFS: fine particles can improve packing at moderate dosage, but excess fines and residual binders can reduce workability and create a weaker interfacial zone [6–12].

The maximum 28-day compressive strength was 73.58 MPa for 12% MK and 20% WFS. This was 22.94% higher than the control strength of 59.85 MPa. At 7 days, the same mixture reached 54.08 MPa, representing an increase of 42.96% over the control value of 37.83 MPa. The absolute maximum 7-day strength, 56.81 MPa, occurred at 12% MK and 30% WFS, equivalent to a 50.17% increase over the control. The shift from 30% WFS at 7 days to 20% WFS at 28 days suggests that early-age densification and later-age hydration did not respond identically to the higher WFS fraction.

Metakaolin exerted the dominant positive effect up to 12% replacement. Its pozzolanic reaction consumed calcium hydroxide and produced additional binding phases, while the microfiller effect refined capillary pores and improved packing around aggregate particles. At 15% MK, strength declined at every WFS level; the 28-day values ranged from 56.80 to 59.74 MPa. The decline is attributed to a combination of cement dilution, increased water demand and possible particle agglomeration. The results therefore indicate that 12% was the practical upper optimum for the present materials and water-to-binder ratio.

Table 4. Seven-day compressive strength matrix (MPa).

MK (%)	WFS 0%	WFS 10%	WFS 20%	WFS 30%	WFS 40%
0	37.83	38.45	40.33	39.00	37.75
3	48.90	49.58	50.82	49.71	48.27
6	47.60	46.46	49.92	50.68	47.14
9	49.20	50.28	51.88	50.47	48.93
12	51.80	52.74	54.08	56.81	50.92
15	43.90	44.83	46.17	45.02	43.19

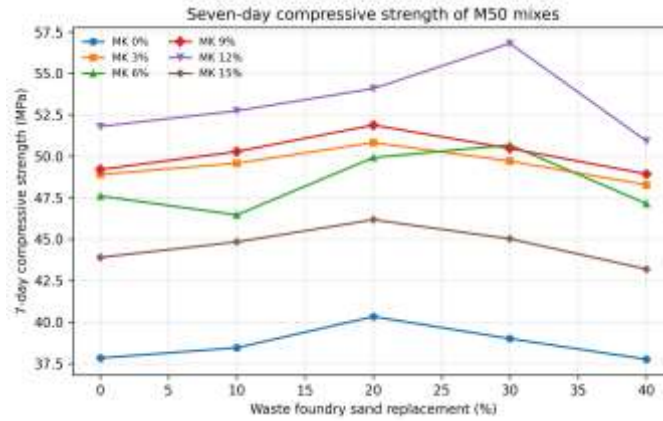


Figure 2. Seven-day compressive strength at different MK and WFS replacement levels.

Table 5. Twenty-eight-day compressive strength matrix (MPa).

MK (%)	WFS 0%	WFS 10%	WFS 20%	WFS 30%	WFS 40%
0	59.85	62.00	61.67	60.33	60.15
3	66.10	67.18	69.42	67.03	66.02
6	64.90	65.98	67.28	69.12	66.33
9	67.80	69.12	70.62	69.18	68.08
12	70.90	72.02	73.58	71.88	70.21
15	56.80	58.18	59.74	58.41	57.09

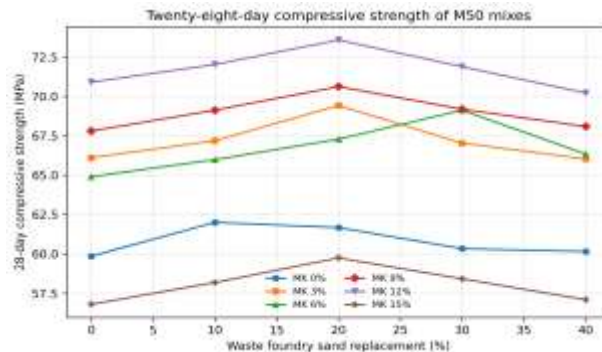


Figure 3. Twenty-eight-day compressive strength at different MK and WFS replacement levels.

3.3 Split tensile and flexural strength

In the WFS-only screening series, 28-day split tensile strength increased from 4.18 MPa for the control to 4.97 MPa at 40% WFS, an improvement of 18.90%. The corresponding 7-day values increased from 3.02 to 3.58 MPa. The continuous increase in this series suggests improved mechanical interlock and particle packing in the tested WFS material, although the compressive-

strength matrix indicates that the most favourable WFS level for combined high-strength performance remained moderate rather than maximum.

The MK-only screening series displayed a clear optimum at 12% replacement. The 28-day split tensile strength increased from 4.18 to 5.12 MPa, a 22.49% improvement. At 15% MK, it fell to 4.02 MPa. Flexural strength followed the same pattern: 12% MK produced 8.68 MPa at 28 days compared with 6.75 MPa for the control, an improvement of 28.59%, whereas 15% MK reduced the value to 6.40 MPa. These trends support the compressive-strength evidence that the beneficial pozzolanic and filler effects were maximized near 12% MK and were offset by dilution and workability penalties at 15%.

For the WFS-only flexural series, the highest 28-day value was 8.10 MPa at 40% WFS, 20.0% above the control. A local reduction occurred at 30% WFS before the maximum at 40%, indicating experimental variability or a non-monotonic response. Because the tensile and flexural tests were conducted as separate one-factor series, the values cannot establish the tensile or flexural response of the combined 12% MK–20% WFS mixture. A future factorial program is required for that purpose.

Table 6. One-factor split tensile and flexural strength results.

Series	Replacement (%)	Split tensile strength (MPa)		Flexural strength (MPa)		Split change at 28 d (%)	Flexural change at 28 d (%)
		7 d	28 d	7 d	28 d		
WFS	0	3.02	4.18	6.30	6.75	0.0	0.0
WFS	10	3.17	4.36	7.05	7.29	4.3	8.0
WFS	20	3.21	4.53	7.65	7.88	8.4	16.7
WFS	30	3.39	4.72	7.73	7.35	12.9	8.9
WFS	40	3.58	4.97	7.62	8.10	18.9	20.0
MK	0	3.02	4.18	6.30	6.75	0.0	0.0
MK	3	3.28	4.61	7.69	8.27	10.3	22.5
MK	6	3.19	4.45	7.42	7.98	6.5	18.2
MK	9	3.42	4.78	7.78	8.37	14.4	24.0
MK	12	3.66	5.12	8.07	8.68	22.5	28.6
MK	15	2.92	4.02	5.95	6.40	-3.8	-5.2

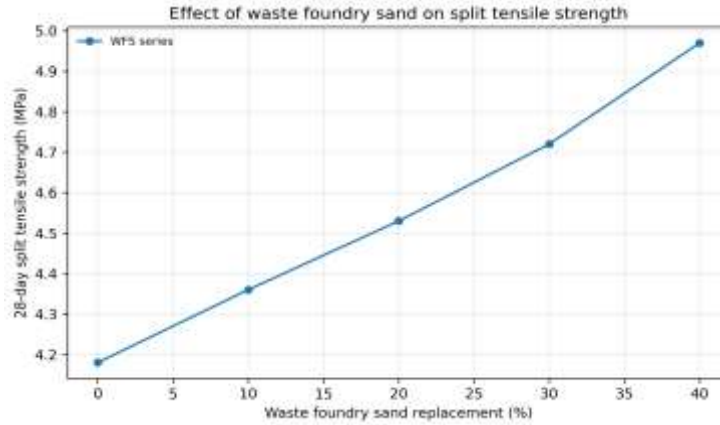


Figure 4. Twenty-eight-day split tensile strength in the WFS screening series.

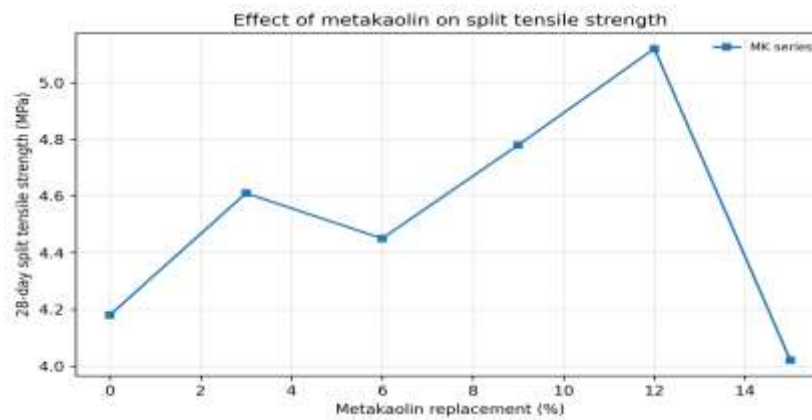


Figure 5. Twenty-eight-day split tensile strength in the MK screening series.

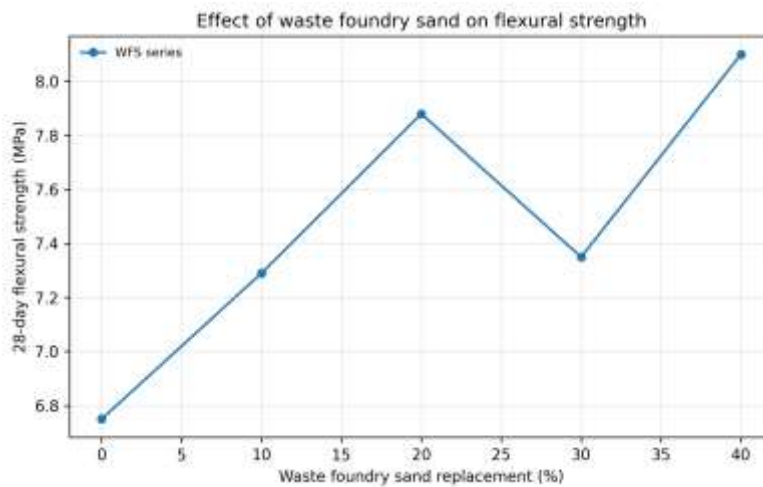


Figure 6. Twenty-eight-day flexural strength in the WFS screening series.

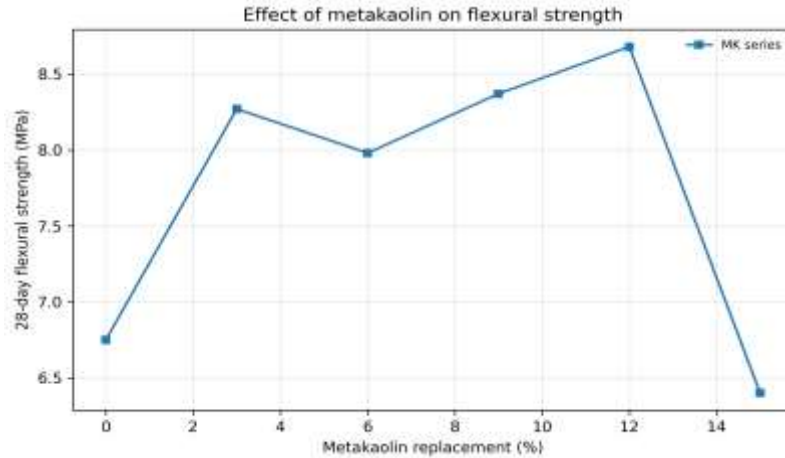


Figure 7. Twenty-eight-day flexural strength in the MK screening series.

3.4 Selection of the balanced optimum

Selection of an optimum mixture should not be based solely on the highest single-age strength. The 12% MK–30% WFS mixture produced the highest 7-day compressive strength, but its slump was 102 mm and its 28-day strength was 71.88 MPa. The 12% MK–20% WFS mixture produced the highest 28-day strength of 73.58 MPa while retaining a slump of 118 mm. It also exceeded the M50 characteristic strength by a substantial margin at both test ages. On this basis, 12% MK with 20% WFS was selected as the balanced optimum for the present material system. The optimum can be explained by complementary microfilling and reaction mechanisms. WFS supplies fine siliceous particles that improve packing and reduce voids at moderate content, while MK reacts with portlandite and refines the binder matrix. At excessive replacement, the demand for paste and water rises, compaction becomes less efficient and the reduction in Portland-cement content is no longer fully compensated by pozzolanic reaction. The optimum therefore represents a balance between densification and dilution.

3.5 Indicative material-cost implication

A preliminary cost comparison was made using the project-assumed rates of ₹8.6/kg for cement, ₹1.8/kg for natural sand, ₹1.6/kg for coarse aggregate, ₹0.6/kg for WFS and ₹25/kg for MK. Water, admixture, processing, transport, quality-control and disposal credits were excluded. On this basis, the control material cost was approximately ₹6,744/m³, while the balanced optimum cost was approximately ₹7,407/m³, an increase of 9.8%. The low cost of WFS partly offset the substantially higher unit cost of MK, but did not eliminate it.

When normalized by 28-day compressive strength, the indicative material cost decreased from approximately ₹112.7 per MPa for the control to ₹100.7 per MPa for the balanced optimum, a reduction of about 10.7%. This strength-normalized comparison is not a complete life-cycle cost analysis, but it indicates that the higher direct material cost was accompanied by a proportionally larger increase in strength. Service-life benefits cannot be claimed without durability testing.

Table 7. Indicative direct material-cost comparison based on project-assumed rates.

Parameter	Control	12% MK + 20% WFS	Difference	Remark
Direct material cost (₹/m ³)	6,744	7,407	+9.8%	Excludes water, SP, transport and processing
28-day compressive strength (MPa)	59.85	73.58	+22.94%	Measured mean
Cost per MPa (₹·m ⁻³ ·MPa ⁻¹)	112.7	100.7	-10.7%	Indicative strength-normalized value
Portland cement (kg/m ³)	434.3	379.6	-54.7	12.6% reduction
Natural sand (kg/m ³)	574.2	452.2	-122.0	21.2% reduction
WFS utilized (kg/m ³)	0	113.0	+113.0	Industrial by-product diverted into concrete

4. LIMITATIONS AND FUTURE WORK

The results have only been obtained with the tested cement and the metakaolin and foundry-sand sources. Variations in the WFS composition are possible due to the metal being cast, the system of binders, and the number of reuses; therefore, characterization and preliminary batching are required prior to field application. The strength was evaluated only at 7 and 28 days. The properties of long-term strength, drying shrinkage, creep, chloride penetration, sulphate resistance, carbonation, water absorption, freeze–thaw performance and reinforcement corrosion were not examined. Lack of leaching test means that environmental suitability must be checked before using the WFS as a structure.

The split tensile and flexural were not a complete MK × WFS matrix, but were one factor screening studies. Future work should then be conducted to test the combined mixtures for tensile, flexural and durability properties and backed up by SEM, XRD and EDS to verify the pore refinement and hydration products. Structural-scale beams, slabs and columns should be checked as well, after which life-cycle assessment and field trials should be conducted.

5. Conclusions

1. Workability was found to decrease with increasing MK and WFS. The slump was found to be in the range of 125 mm (control) to 90 mm (15% MK & 40% WFS). Slump of the selected mixture (12% MK–20% WFS) was found to be satisfactory (118 mm).
2. The maximum compressive strength at 12% MK and 20% WFS was obtained after 28 days, which is 73.58 MPa, increasing by 22.94 % compared to the control value of 59.85 MPa.
3. For the same balanced optimum, the 7 days strength was 42.96% higher at 54.08 MPa. The absolute maximum 7-day strength was 56.81 MPa at 12% MK and 30% WFS.
4. The compressive strength of the WFS replacement was improved with moderate replacement but optimum content of WFS ranged from 10% to 30% for different levels of MKs and different curing ages. The best overall level was 20% WFS for 28 days of performance.
5. The use of metakaolin up to 12% replacement level enhanced all strength parameters measured. A rise in MK to 15% resulted in a clear decrease, which was due to cement dilution, water demand, and possible agglomeration.
6. Compared to the control, 12% MK resulted in a 22.49% and 28.59% increase in 28-days split tensile strength and flexural strength, respectively, in separate screening series.
7. The optimal balanced design resulted in a reduction of the Portland cement content by 54.7 kg/m³ and the use of 113.0 kg/m³ of WFS, as well as a decrease in the use of natural sand by 122.0 kg/m³.
8. The project assumed cost rates of direct materials increased by around 9.8 % and the indicative cost per unit of 28-day strength decreased by around 10.7 %. Before making economic claims for service life, a full life-cycle/durability assessment must be performed.

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